LIE ALGEBRA COHOMOLOGY AND SEIBERG-WITTEN THEORY

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ABSTRACT. We compute the algebra of perturbative holomorphic-topological local operators of pure $\mathcal{N} = 2$ gauge theory with gauge group $SU(2)$, proving a conjecture of the author: it is the Poisson vertex algebra generated by a Virasoro element X of central charge zero and an odd primary Y of weight three, subject only to $X^2 = 0$. Equivalently, we compute the relative Lie algebra cohomology $H^*(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2)$ and the self-Ext groups of the unit in the coherent Satake category for SL_2. The proof uses factorization on configuration spaces of points on the affine line, a depth argument, and two deformations of the coefficient algebra $\mathbb{C}[z,w]/(w^2)$.

1. INTRODUCTION

Much of the appeal of supersymmetric quantum field theories comes from the ability to calculate non-trivial quantities exactly. Examples include the Witten index, the elliptic genus, the superconformal index and the Schur index. A deeper study often reveals that these indices are just the shadow of a much richer invariant, carrying interesting and non-trivial algebraic structure. For example, the Schur index of a four-dimensional $\mathcal{N} = 2$ superconformal field theory is the shadow of a chiral algebra [BLL+15]. Studying the spaces underlying these indices also reveals that such "superconformal" indices are in fact well-defined in the more general non-conformal setting: they are characters of the space of local operators in a mixed holomorphic-topological twist [Kap06, K26].

This paper studies the space of local operators in the holomorphic-topological twist of the pure four-dimensional $\mathcal{N} = 2$ gauge theory with gauge group $SU(2)$ [SW94]. In a previous paper [K26], the author proposed the following conjecture: to all orders in perturbation theory, the space of local operators is given by the Poisson vertex algebra A, generated by an even field X and an odd field Y with λ-brackets

$$\{X_\lambda X\} = \partial X + 2\lambda X, \qquad \{X_\lambda Y\} = \partial Y + 3\lambda Y, \qquad \{Y_\lambda Y\} = 0, \tag{1.1}$$

divided by the smallest Poisson vertex ideal containing X^2. In this paper we prove that conjecture.

To state the result more precisely, we recall from [K26] that the space of local operators of the twisted $SU(2)$ theory is given, to all orders in perturbation theory [ESW22, BG25], by the relative Lie algebra cohomology

$$\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2) = H^*\big(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2\big), \tag{1.2}$$

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where $\mathfrak{sl}_2[z,w]/(w^2)$ denotes the infinite-dimensional Lie algebra $\mathfrak{sl}_2 \otimes \mathbb{C}[z,w]/(w^2)$ and w denotes an even square-zero variable (written $\varepsilon = dz$ in [K26]). There we showed that this cohomology carries a Poisson vertex algebra structure, and constructed a homomorphism of Poisson vertex algebras $\varphi : A \to \mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$.

Theorem 1.1. *The map $\varphi : A \to \mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$ is an isomorphism of Poisson vertex algebras. In particular,*

$$H^*\big(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2\big) \;\cong\; \frac{\mathbb{C}[X, \partial X, \ldots] \otimes \Lambda[Y, \partial Y, \ldots]}{\langle X^2,\ XY,\ X\partial Y,\ Y\partial Y\rangle_\partial}$$

as differential supercommutative algebras graded by spin, ghost number and B-number.

Thus every perturbative local operator of the twisted $\mathcal{N} = 2$ theory is a differential polynomial in the two gaugino composites

$$X = -\operatorname{tr}(\bar\lambda\lambda), \qquad Y = \tfrac{1}{3}\operatorname{tr}(\bar\lambda^3),$$

where λ and $\bar\lambda$ are the gaugino components, one of each chirality, that survive the holomorphic-topological twist, and all relations among them follow from $X^2 = 0$ and the axioms of a Poisson vertex algebra.

Recall that $\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$ is graded by ghost number, spin and B-number, which in terms of (1.2) are the cohomological degree and the gradings induced by $\deg z = \deg w = -1$ and by $\deg z = 0$, $\deg w = -1$, respectively; each trigraded piece is finite-dimensional. Theorem 1.1 determines these dimensions, and combined with the computation of the Hilbert–Poincaré series of A in [K26, (2.65)], it gives the following.

Corollary 1.2. *Let t, q and y keep track of the ghost number, the spin and the B-number. Then*

$$P_{\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)}(t,q,y) = \sum_{n\geq 0} q^{n(n+1)}\, t^{2n} y^n\, \frac{(-tqy^2;q)_n}{(q;q)_n}.$$

Specializing to $t = -1$ gives the Macdonald index of the pure $\mathcal{N} = 2$ theory,

$$\mathcal{I}_{\mathrm{M}}(q,y) = \sum_{n\geq 0} q^{n(n+1)} y^n\, \frac{(qy^2;q)_n}{(q;q)_n}, \tag{1.3}$$

and further setting $y = 1$ gives its Schur index,

$$\mathcal{I}_{\mathrm{S}}(q) = \sum_{n\geq 0} q^{n(n+1)}. \tag{1.4}$$

To our knowledge, the formula (1.3) for the Macdonald index of the pure $SU(2)$ theory is new.[1]

The Macdonald index is also given by a Molien-type integral over the maximal torus of $SU(2)$. Comparing this integral with (1.3) proves a new constant term identity, predicted in [K26, (3.127)]:[2]

$$\mathrm{CT}_z\Big[\tfrac{1}{2}(1-z^2)(1-z^{-2})\,(q;q)_\infty (qz^{\pm 2};q)_\infty\,(qy;q)_\infty (qyz^{\pm 2};q)_\infty\Big] = \sum_{n\geq 0} q^{n(n+1)} y^n\, \frac{(qy^2;q)_n}{(q;q)_n}. \tag{1.5}$$

[1] In [Son16], a different expression for the Macdonald index was proposed on the basis of a 2d TQFT description of the theory, and checked to low order in q.

[2] Here CT_z denotes the constant term in z, and $(qx^{\pm 1};q)_\infty := (qx;q)_\infty (qx^{-1};q)_\infty$.

1.1. **Outline of the proof.** Injectivity of φ is the easier half of Theorem 1.1. In §4 we construct, for every element of the graded dual of A, an explicit Chevalley–Eilenberg cycle that detects it. The cycles live on the Cartan current algebra $\mathfrak{h}[z] \ltimes w\,\mathfrak{sl}_2[z]$, where the chain complex becomes a Koszul complex, and a change of variables from power sums to elementary symmetric functions produces them in closed form.

The main difficulty is surjectivity: one has to show that the relative cohomology contains nothing beyond $\varphi(A)$. Our argument is algebro-geometric and goes as follows.

Fix the B-number B, that is, the number of b-ghosts. The complex $\mathrm{Obs}_{\mathrm{HT}}(\mathfrak{sl}_2)$ comes equipped with derivations T_m, $m \geq 1$, which lower the mode of every b-ghost by m and annihilate the c-ghosts, commute with the BRST differential. On the graded dual of the cohomology they act as the power sums $\sum_i t_i^m$ of B commuting variables $t_1, \dots, t_B$, which we think of as the *positions* of the b-ghosts on the z-line. In this way the dual of the cohomology at B-number B becomes a coherent sheaf $\mathcal{K}_B$, graded by the number p of c-ghosts, on the configuration space

$$\mathrm{Sym}^B \mathbb{A}^1 = \{\text{effective divisors } t_1 + \cdots + t_B \text{ on the } z\text{-line}\} \;\cong\; \mathbb{A}^B.$$

The point of this reformulation is that the relative BGG resolution presents $\mathcal{K}_B$ as the homology of a complex of vector bundles of length B on $\mathrm{Sym}^B \mathbb{A}^1$ (§5).

The algebra A defines a second sheaf on the same space. The generator $X \mapsto -\sum_a b^a \partial c^a$ carries a single b-ghost: it is a *simple point*. The generator $Y \mapsto b^1 b^2 b^3$ carries three b-ghosts at the same location: it is a *triple point*. Accordingly, the part of A with r factors of X and s factors of Y is dual to the pushforward of the structure sheaf along the finite map

$$\nu_{r,s} : \mathrm{Sym}^r \mathbb{A}^1 \times \mathrm{Sym}^s \mathbb{A}^1 \longrightarrow \mathrm{Sym}^{r+3s} \mathbb{A}^1, \qquad (D, D') \longmapsto D + 3D',$$

which is the normalization of the closure of the stratum of divisors with r simple and s triple points; the relations $X^2 = XY = X\partial Y = Y\partial Y = 0$ become exactly the vanishing orders along diagonals that carve out this normalization (Lemma 4.1). Injectivity of φ gives a surjection of sheaves

$$\psi_B : \mathcal{K}_B \twoheadrightarrow \mathcal{A}_B^\vee := \bigoplus_{r+3s=B} \nu_{r,s*}\mathcal{O},$$

and Theorem 1.1 asserts that its kernel, the *defect*, vanishes. We prove this in three steps, which we call *factorization*, *depth* and *deformation*; see Figure 1.

Factorization. Over the open subset of $\mathrm{Sym}^B \mathbb{A}^1$ where a divisor splits as $D_1 + D_2$ with disjoint supports, both sheaves factorize:

$$\mathcal{K}_B \cong \mathcal{K}_{B_1} \boxtimes \mathcal{K}_{B_2}, \qquad \mathcal{A}_B^\vee \cong \mathcal{A}_{B_1}^\vee \boxtimes \mathcal{A}_{B_2}^\vee.$$

For $\mathcal{A}^\vee$ this is the Chinese remainder theorem. For $\mathcal{K}$ it is a Lie-algebraic form of locality: the cohomology of $\mathfrak{sl}_2[z]$ with coefficients supported at two disjoint finite sets of points is the tensor product of the two cohomologies. We deduce this from Teleman's theorem by an explicit cocycle computation (§6); the resulting structure is reminiscent of the factorization structures of chiral

algebras [BD04]. By induction on B, the map ψ_B is therefore an isomorphism away from the *small diagonal*

$$Z_B = \{B \cdot a : \ a \in \mathbb{A}^1\} \subset \mathrm{Sym}^B \mathbb{A}^1,$$

where all B ghosts collide. Hence the defect is supported on Z_B, a line of codimension $B-1$.

Depth. By the acyclicity lemma of Peskine and Szpiro [PS73] (see also [BE73]), the homology of a short complex of vector bundles cannot contain a nonzero subsheaf supported in high codimension, provided the quotient by it is sufficiently deep there. In our situation the numerology is exactly right. The top term of the BGG complex, which corresponds to the pure-X sector, is a line bundle and splits off by a rank argument. What remains has length $B-1$, exactly the codimension of Z_B, while $\mathcal{A}_B^\vee$ is deep along Z_B: the preimage of Z_B under $\nu_{r,s}$ is a line in the $(r+s)$-dimensional space $\mathrm{Sym}^r \mathbb{A}^1 \times \mathrm{Sym}^s \mathbb{A}^1$, so its codimension $r+s-1$ is at least the number $p=r$ of c-ghosts as soon as $s \geq 1$. A depth count then forces the defect to vanish in every degree $p \geq 2$ (§7).

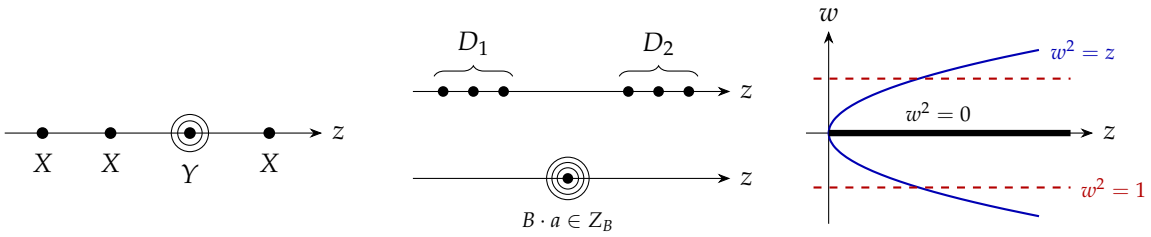
Deformation. The depth bounds give no information in the two lowest degrees $p = 0, 1$. There we deform the coefficient algebra: in the (z,w)-plane, the double line $w^2 = 0$ deforms into the parabola $w^2 = z$ and into the pair of lines $w^2 = 1$,

$$\mathbb{C}[z,w]/(w^2 - z) \cong \mathbb{C}[u], \qquad \mathbb{C}[z,w]/(w^2-1) \cong \mathbb{C}[z] \times \mathbb{C}[z].$$

For both deformed algebras the relative cohomology follows from the Garland–Lepowsky theorem and it is tiny: $\mathbb{C}$ and $\mathbb{C} \oplus \mathbb{C}\,Y$, respectively. The corresponding spectral sequences show that an excess class in degree $p=1$ and B-number B would have to cancel against an excess class in degree $p=0$ and B-number $B+2$, through a map which raises the spin by one for the parabola and by two for the pair of lines. At the lowest spin carrying an excess class these two requirements are incompatible (§8). The algebra A does not interfere, since both deformations act on it with explicit contracting homotopies.

Combining the three steps proves Theorem 1.1 by induction on B (§9).

1.2. **Relation to the affine Grassmannian.** There is a second, more geometric way to think about Theorem 1.1, in terms of coherent sheaves on the affine Grassmannian. Let G be SL_2 or PGL_2, let $\mathbf{O} = \mathbb{C}[[z]]$ and $\mathbf{K} = \mathbb{C}((z))$, and let $\mathrm{Gr}_G = G(\mathbf{K})/G(\mathbf{O})$ be the affine Grassmannian. Following a proposal of Costello reported in [CW19], the category of line operators in the holomorphic-topological twist [Kap06] of pure $\mathcal{N}=2$ gauge theory is the derived category of $G(\mathbf{O})$-equivariant coherent sheaves on Gr_G, made monoidal by convolution, which models the collision of line operators [KS09]. With respect to the perverse coherent t-structure [AB10, BFM05], its heart is the *coherent Satake category* of Cautis and Williams [CW19]. Unlike its constructible counterpart in the geometric Satake correspondence [MV07], this category is neither semisimple nor symmetric monoidal; its simple objects correspond to Wilson–'t Hooft lines, its Grothendieck ring is the quantized K-theoretic Coulomb branch of the theory [BFM05, BFN18], and for GL_n it categorifies a quantum cluster algebra [CW19]. The construction extends to gauge theories with matter [CW23].



(a) ghosts as points (b) separation and collision (c) deformation

FIGURE 1. The three steps of the proof. (a) A cohomology class of B-number B is a function of the positions of B ghosts on the z-line; the generator X is a simple point and Y is a triple point. (b) When the ghosts form separated clusters the cohomology factorizes, so any defect is supported on the small diagonal Z_B, where all ghosts collide; there it is removed by a depth argument. (c) In the two lowest degrees the double line $w^2 = 0$ is deformed into the parabola $w^2 = z$ and into the pair of lines $w^2 = 1$.

In any such description, bulk local operators are the derived endomorphisms of the identity line operator, that is, of the monoidal unit, which is the skyscraper sheaf $\mathcal{O}_e$ at the base point $e = [G(\mathbf{O})]$. Niu identified this endomorphism complex with the basic bc-complex [Niu22, Theorem 4.13]. The tangent space of Gr_G at e is $T_e\mathrm{Gr}_G = \mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O})$, so the self-Ext of $\mathcal{O}_e$ on the formal neighbourhood of e is the exterior algebra $\Lambda^\bullet(\mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O}))$, and passing to derived $G(\mathbf{O})$-invariants adds the Chevalley–Eilenberg cochains of $z\mathfrak{g}(\mathbf{O})$. The result is exactly the basic bc-complex: under the residue pairing $\mathfrak{g}(\mathbf{K})/\mathfrak{g}(\mathbf{O}) \cong (\mathfrak{g}(\mathbf{O})\,dz)^\vee$, the b-ghosts become tangent vectors to Gr_G at e, and the c-ghosts become the generators of the $G(\mathbf{O})$-equivariant cohomology. Since this complex depends on G only through $\mathfrak{g} = \mathfrak{sl}_2$, Theorem 1.1 consequently computes the derived endomorphisms of the unit of the coherent Satake category, for both SL_2 and PGL_2. Here we use the model of [Niu22, §4], in which Ext is taken among $\widetilde{G}(\mathbf{O})$-equivariant ind-coherent sheaves on Gr_G, where $\widetilde{G}(\mathbf{O}) = G(\mathbf{O}) \rtimes \mathbb{C}^\times$ and $\mathbb{C}^\times$ acts through a cover of loop rotation, so that Ext groups are graded by the weight of loop rotation.

Corollary 1.3. *Let $G = SL_2$ or PGL_2. There is an isomorphism of trigraded vector spaces*

$$\mathrm{Ext}^\bullet_{\widetilde{G}(\mathbf{O})}(\mathcal{O}_e, \mathcal{O}_e) \;\cong\; H^*\big(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2\big) \;\cong\; A,$$

under which the Ext *degree, the polyvector degree and the weight of loop rotation correspond to the ghost number, the B-number and the spin, respectively. In particular, the Poincaré series of* $\mathrm{Ext}^\bullet_{\widetilde{G}(\mathbf{O})}(\mathcal{O}_e, \mathcal{O}_e)$ *is given by Corollary 1.2.*

Under this identification, Y is the invariant trivector on the first-order normal directions $\mathfrak{g}\,z^{-1} \subset T_e\mathrm{Gr}_G$, and X pairs these directions with the first-order directions $z\mathfrak{g}$ of $G(\mathbf{O})$. Moreover, the derivations T_m of §1.1 act on $T_e\mathrm{Gr}_G$ as multiplication by z^m, so the factorization of §6 is a Lie-algebraic shadow of the factorization of the Beilinson–Drinfeld Grassmannian over configurations of points [BD04, MV07].

The quasi-isomorphism of [Niu22] is stated for complexes. One expects it to intertwine the Yoneda product with the product of the basic complex, in which case Corollary 1.3 is an isomorphism of algebras, and the E_2-structure on the endomorphisms of the unit to induce the Poisson vertex algebra structure of $\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$, with the λ-bracket arising as the secondary product of [BBB+20].

From this point of view Theorem 1.1 is perhaps surprising. In the constructible equivariant Satake category the derived endomorphism algebra of the unit is just $H^*_G(\mathrm{pt})$, a polynomial ring (cf. [BF08]); for coherent sheaves it is the infinitely generated algebra A, whose relations are generated by $X^2 = 0$. Niu showed that the graded Euler characteristic of this complex is the Schur index [Niu22], but its cohomology was not determined there; Theorem 1.1 computes it, together with its refinement by B-number. Since the endomorphisms of the unit of a monoidal category act on all of its morphism spaces, A also acts on the spaces of junctions between line operators, such as the junctions between Wilson and 't Hooft lines. The study of these Wilson–'t Hooft lines will be the subject of a subsequent paper.

1.3. **Relation to other work.** The cohomology of current algebras with nilpotent coefficients has a long history. The strong Macdonald conjecture of Hanlon and Feigin [Han90, Fei88], proved by Fishel, Grojnowski and Teleman [FGT08], determines the cohomology of the truncated current algebras $\mathfrak{g}[z]/(z^N)$ and of the current superalgebra $\mathfrak{g}[z,s]$ in which s is an *odd* square-zero variable. The Lie algebra $\mathfrak{sl}_2[z,w]/(w^2)$ differs from $\mathfrak{sl}_2[z,s]$ only in the parity of the square-zero variable, so it is natural to ask whether these techniques apply to it. The answer, however, is of a different nature: the cohomology rings computed in [FGT08] are free, whereas the algebra A has infinitely many relations. Accordingly, our method is different: instead of harmonic representatives, we use the module structure over symmetric functions, factorization and depth. Factorization at disjoint points of the cohomology of current algebras is also a central theme of Teleman's work on Lie algebra cohomology and the fusion rules [Tel95, Tel96].

1.4. **Organization of the paper.** The paper is organized as follows. In §2 we recall the algebra A, the basic complex and the map φ, and reformulate Theorem 1.1 in dual form. Section 3 collects the representation-theoretic input. Section 4 proves injectivity. Section 5 sets up the sheaves on $\mathrm{Sym}^B \mathbb{A}^1$ and treats the top degree. Sections 6, 7 and 8 carry out separation, collision and deformation, respectively, and §9 assembles the proof and derives the corollaries.

1.5. **Notation and conventions.** All vector spaces, algebras and Lie algebras are over $\mathbb{C}$. For a commutative $\mathbb{C}$-algebra R and a Lie algebra $\mathfrak{g}$ we write $\mathfrak{g} \otimes R$ for the current algebra, and for an ideal $(f) \subset \mathbb{C}[z,w]$ we abbreviate $\mathfrak{g}[z,w]/(f) := \mathfrak{g} \otimes \mathbb{C}[z,w]/(f)$. For a graded vector space $W = \bigoplus_S W_S$ with finite-dimensional pieces we write $W^\vee = \bigoplus_S W_S^*$ for its graded dual. The conventions for cochains of current algebras, polynomial in the modes or with finite-dimensional coefficients, are fixed in §2.2 and at the beginning of §3.

1.6. **Use of AI tools.** Many of the proof techniques in this paper were suggested by the large language models GPT-6 Astra and Claude Opus 5.5. Claude Opus 5.5 also helped with the writing and proofreading of the manuscript. Under the author's direction, the two models were also coordinated to produce a complete Lean formalization of Theorem 1.1, available in a companion repository.[3] The author takes full responsibility for the content of this paper, including any errors.

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2. THE ALGEBRA A AND THE BASIC COMPLEX

2.1. **The Poisson vertex algebra A.** Let $V = \mathbb{C}[X, \partial X, \ldots] \otimes \Lambda[Y, \partial Y, \ldots]$ be the free Poisson vertex algebra with λ-brackets (1.1), and let $I \subset V$ be the smallest Poisson vertex ideal containing X^2, so that $A = V/I$. A routine computation with the λ-brackets shows that I is the differential ideal generated by X^2, XY, $X\partial Y$ and $Y\partial Y$, which gives the presentation of A in Theorem 1.1.

The algebra A carries three gradings: X and Y have spin 2 and 3, ghost number 2 and 3, and B-number 1 and 3, and ∂ raises the spin by one. It is convenient to replace the ghost number gh by $p := \mathrm{gh} - B$, so that

$$(B, p, S)(\partial^k X) = (1, 1, k+2), \qquad (B, p, S)(\partial^k Y) = (3, 0, k+3). \tag{2.1}$$

We write S for the spin, since the letter s will count factors of Y. The ideal I is homogeneous for the number of X's and the number of Y's, so $A = \bigoplus_{r,s\geq 0} A_{r,s}$, where $A_{r,s}$ is spanned by the images of monomials with r factors of the form $\partial^k X$ and s factors of the form $\partial^k Y$. By (2.1), $A_{r,s}$ lies in $(B, p) = (r + 3s, r)$. Since (B, p) determines (r, s), we write

$$A^p_B := A_{r,s} \text{ if } (B, p) = (r + 3s, r), \qquad A^p_B := 0 \text{ if } (B, p) \text{ is not of this form.} \tag{2.2}$$

2.2. **The basic complex.** We follow the conventions of [K26, §3] for the bc-system and its BRST differential. Thus we identify $\mathfrak{sl}_2 \cong \mathfrak{so}_3$, with basis e_1, e_2, e_3, $[e_a, e_b] = \epsilon_{abc} e_c$, and invariant form δ_{ab}; repeated color indices are summed. In terms of the ghost fields b^a and c^a we use the modes

$$b^a_j := \tfrac{1}{j!}\partial^j b^a \quad (j \geq 0), \qquad c^a_n := \tfrac{1}{n!}\partial^n c^a \quad (n \geq 1), \tag{2.3}$$

all of which are odd. The basic complex is the invariant part

$$\mathcal{C} := \Lambda\big(b^a_j,\, c^a_n :\ j \geq 0,\ n \geq 1,\ a = 1, 2, 3\big)^{\mathfrak{sl}_2} \tag{2.4}$$

[3] https://github.com/qqz3ro/pure-su2-pva

of the exterior algebra on these modes (finite polynomial expressions only), and the BRST differential reads in modes

$$Qc_n^a = \frac{1}{2}\sum_{i=1}^{n-1} \epsilon_{abc}\, c_i^b c_{n-i}^c, \qquad Qb_j^a = \sum_{i=1}^{j} \epsilon_{abc}\, c_i^b b_{j-i}^c. \tag{2.5}$$

The gradings are $B = \#b$, $p = \#c$, and the spin $S(b_j^a) = j+1$, $S(c_n^a) = n$; the ghost number is $\mathrm{gh} = B + p$. The differential has degree $(0,1,0)$ in (B,p,S), and each component of fixed (B,p,S) is finite-dimensional. We put

$$H_B^p := H^p(\mathcal{C}_B, Q), \qquad H := \bigoplus_{B,p} H_B^p = \mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2). \tag{2.6}$$

Lie algebra cohomology. The algebra $\mathfrak{sl}_2[z,w]/(w^2) = \mathfrak{sl}_2[z] \ltimes w\,\mathfrak{sl}_2[z]$ has the basis of modes $e_a z^n$ and $we_a z^j$, and c_n^a and b_j^a are the coordinates dual to the modes with $n \geq 1$ and $j \geq 0$. Then Q is the Chevalley–Eilenberg differential of $\mathfrak{sl}_2[z,w]/(w^2)$, restricted to cochains that are basic for the constant subalgebra $\mathfrak{sl}_2 = \mathfrak{sl}_2 \otimes 1$: horizontal (no c_0^a) and invariant. We therefore write

$$H^*\big(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2\big) := H^*(\mathcal{C}, Q), \tag{2.7}$$

the cohomology of basic cochains which are polynomial in the modes. The B-number is the w-degree, and $H_B^p = H^p\big(\mathfrak{sl}_2[z], \mathfrak{sl}_2; \Lambda^B((\mathfrak{sl}_2[z]\,dz)^\vee)\big)$ with graded duals; this is the precise meaning of (1.2). Since each graded piece is finite-dimensional, this convention is the natural one, and it is part of the statement of Theorem 1.1. We use the same notation $H^*(\mathfrak{sl}_2 \otimes R, \mathfrak{sl}_2)$ in §8 for $R = \mathbb{C}[z,w]/(w^2 - z)$ and $R = \mathbb{C}[z,w]/(w^2-1)$, which have the same basis of modes z^n, wz^j.

Translations and positions. In modes, the holomorphic derivative is the even derivation

$$\partial b_j^a = (j+1)\, b_{j+1}^a, \qquad \partial c_n^a = (n+1)\, c_{n+1}^a. \tag{2.8}$$

A direct computation from (2.5) gives $[Q, \partial] = -\sum_a c_1^a L_a$, where L_a is the derivation by which the constant mode e_a acts on the coordinates, $L_a b_j^c = \epsilon_{acd} b_j^d$ and $L_a c_n^c = \epsilon_{acd} c_n^d$. Hence ∂ commutes with Q on $\mathcal{C}$ and acts on H; it is the translation of the Poisson vertex algebra $\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$. For $m \geq 1$ we also use the even derivations

$$T_m b_j^a = b_{j-m}^a \quad (\text{zero if } j < m), \qquad T_m c_n^a = 0. \tag{2.9}$$

They preserve $\mathcal{C}$ and commute with Q: both $T_m Q$ and $Q T_m$ vanish on the c's, and on b_j^a both give $\sum_{i=1}^{j-m} \epsilon_{abc} c_i^b b_{j-m-i}^c$. The derivation T_m lowers the mode of each b-ghost by m; it is the transpose of multiplication by z^m on the coefficient module $w\,\mathfrak{sl}_2[z]$. In §5.2 it will measure the positions of the b-ghosts.

2.3. **The map φ.** Put

$$\mathsf{X} := -\sum_a b_0^a c_1^a, \qquad \mathsf{Y} := b_0^1 b_0^2 b_0^3 = \tfrac{1}{6}\,\epsilon_{abc}\, b_0^a b_0^b b_0^c. \tag{2.10}$$

These are the gaugino composites of the introduction. Indeed, under the identification $b^a \leftrightarrow \bar\lambda^a$, $\partial c^a \leftrightarrow \lambda^a$ of [K26, Remark 3.3], with $\bar\lambda = \sum_a \bar\lambda^a e_a$ and $\lambda = \sum_a \lambda^a e_a$, we have $\mathsf{X} = -\operatorname{tr}(\bar\lambda\lambda)$ and

$\mathsf{Y} = \frac{1}{3}\operatorname{tr}(\bar\lambda^3)$, where tr is the trace in any representation of $\mathfrak{sl}_2$, rescaled so that $\operatorname{tr}(e_a e_b) = \delta_{ab}$. For the second identity, note that $\bar\lambda^2 = \frac{1}{2}[\bar\lambda,\bar\lambda]$ because $\bar\lambda$ is odd, so that $\operatorname{tr}(\bar\lambda^3) = \frac{1}{2}\epsilon_{abc}\bar\lambda^a\bar\lambda^b\bar\lambda^c$. Since $Y \mapsto \lambda Y$ is an automorphism of A for every $\lambda \in \mathbb{C}^\times$ (the relations and the λ-brackets are homogeneous in Y), the normalization of Y, which differs from that of [K26] by a factor $1/6$, is immaterial for Theorem 1.1.

Lemma 2.1. *The cochains* X *and* Y *are Q-closed, and*

$$\mathsf{X}^2 = Q\big(-\epsilon_{abc}\, b_0^a b_0^b c_2^c\big), \qquad \mathsf{X}\mathsf{Y} = 0, \qquad \mathsf{X}\,\partial\mathsf{Y} = Q\Big(-\tfrac{1}{4}\big(\textstyle\sum_a b_0^a b_1^a\big)^2\Big), \qquad \mathsf{Y}\,\partial\mathsf{Y} = 0. \tag{2.11}$$

Proof. By (2.5), $Qb_0^a = Qc_1^a = 0$. Every term of $\mathsf{X}\mathsf{Y}$ and of $\mathsf{Y}\,\partial\mathsf{Y}$ contains a repeated b_0^a. Using $Qc_2^c = \frac{1}{2}\epsilon_{cde}c_1^d c_1^e$ and $\epsilon_{abc}\epsilon_{cde} = \delta_{ad}\delta_{be} - \delta_{ae}\delta_{bd}$ we find $Q(\epsilon_{abc}b_0^a b_0^b c_2^c) = b_0^a b_0^b c_1^a c_1^b = -\mathsf{X}^2$. Similarly, $\sigma := \sum_a b_0^a b_1^a$ satisfies $Q\sigma = -\epsilon_{abc}b_0^a b_0^b c_1^c$, hence $Q(\sigma^2) = 2\sigma\, Q\sigma = -4\,\mathsf{Y}\sum_a b_1^a c_1^a = -4\,\mathsf{X}\,\partial\mathsf{Y}$. □

Consequently, the assignment $X \mapsto [\mathsf{X}]$, $Y \mapsto [\mathsf{Y}]$, extended multiplicatively and ∂-equivariantly, defines a homomorphism of differential algebras

$$\varphi : A \longrightarrow H.$$

Up to the normalization of Y, this is the map of [K26, §3.6]. In particular it intertwines the λ-brackets (1.1) with the Poisson vertex algebra structure of $\mathsf{Obs}_{\mathsf{HT}}(\mathfrak{sl}_2)$, which is induced from the bc-system with $\{b^a{}_\lambda c^b\} = \delta_{ab}$. Indeed, writing $b^a = b_0^a$ and $\partial c^a = c_1^a$, one finds $\{\mathsf{X}_\lambda b^a\} = (\partial+\lambda)b^a$ and $\{\mathsf{X}_\lambda \partial c^a\} = (\partial+\lambda)\partial c^a$, and the Leibniz rule gives

$$\{\mathsf{X}_\lambda \mathsf{X}\} = (\partial + 2\lambda)\mathsf{X}, \qquad \{\mathsf{X}_\lambda \mathsf{Y}\} = (\partial + 3\lambda)\mathsf{Y}, \qquad \{\mathsf{Y}_\lambda \mathsf{Y}\} = 0.$$

Hence the map from the free algebra V preserves λ-brackets and kills X^2, so its kernel is a Poisson vertex ideal containing X^2, and therefore contains I. Lemma 2.1 makes the vanishing on the generators of I explicit. The map φ preserves the gradings (2.1), so Theorem 1.1 is equivalent to the bijectivity of

$$\varphi : A_B^p \longrightarrow H_B^p \qquad \text{for all } (B,p).$$

2.4. **Dual formulation.** With graded duals as in §1.5, put

$$K_{B,p} := (H_B^p)^\vee, \qquad A_{B,p}^\vee := (A_B^p)^\vee, \qquad \psi_{B,p} := \varphi^\vee : K_{B,p} \longrightarrow A_{B,p}^\vee. \tag{2.12}$$

Theorem 1.1 is equivalent to the statement that every $\psi_{B,p}$ is an isomorphism. We show that every $\psi_{B,p}$ is surjective in §4, and that it is injective by induction on B in §§5–9.

3. Representation-theoretic input

Throughout, $\mathfrak{l} := \mathfrak{sl}_2[z]$, V_m is the irreducible $\mathfrak{sl}_2$-module of highest weight m (of dimension $m+1$), and for $a \in \mathbb{C}$, $V_m(a)$ is the evaluation module of $\mathfrak{l}$ on which $x\,f(z)$ acts by $f(a)\,x$. For an $\mathfrak{l}$-module M we write

$$C_*(\mathfrak{l},\mathfrak{sl}_2;M) = \big(\Lambda^\bullet(\mathfrak{l}/\mathfrak{sl}_2)\otimes M\big)_{\mathfrak{sl}_2}, \qquad C^*(\mathfrak{l},\mathfrak{sl}_2;M) = \operatorname{Hom}_{\mathfrak{sl}_2}\big(\Lambda^\bullet(\mathfrak{l}/\mathfrak{sl}_2), M\big)$$

for the relative chains and cochains. When M is finite-dimensional we use the full Hom, that is, ordinary algebraic cochains; when M is graded with finite-dimensional graded pieces, as in §2.2, we use graded duals.

Theorem 3.1 (Relative BGG resolution [GL76, Lep79]). *The trivial $\mathfrak{l}$-module has a resolution by $(\mathfrak{l}, \mathfrak{sl}_2)$-relatively projective modules*

$$\cdots \to P_2 \to P_1 \to P_0 \to \mathbb{C} \to 0, \qquad P_p = U(\mathfrak{l}) \otimes_{U(\mathfrak{sl}_2)} V_{2p}[d_p], \qquad d_p = \tfrac{p(p+1)}{2},$$

where $[d_p]$ shifts the polynomial-mode degree by d_p. Consequently, if the constant subalgebra $\mathfrak{sl}_2$ acts locally finitely on an $\mathfrak{l}$-module M, then $H_*(\mathfrak{l}, \mathfrak{sl}_2; M)$ and $H^*(\mathfrak{l}, \mathfrak{sl}_2; M)$ are computed by complexes with terms

$$(M \otimes V_{2p})^{\mathfrak{sl}_2}[d_p] \qquad \text{and} \qquad \mathrm{Hom}_{\mathfrak{sl}_2}(V_{2p}, M)[-d_p], \qquad p \geq 0,$$

respectively.

The underlying computation $H_p(z\,\mathfrak{sl}_2[z]) \cong V_{2p}[d_p]$ is the affine A_1 case of the theorem of Garland and Lepowsky [GL76], the affine analogue of Kostant's theorem [Kos61]. The resolution by generalized Verma modules, together with the resulting complex computing relative (co)homology, is due to Lepowsky [Lep79, Proposition 5.3, Corollary 5.4 and Theorem 5.5]; we use it after exchanging positive and negative currents, which reverses the sign of the mode grading. See also [Kum02]. Since $V_{2p}^{\mathfrak{sl}_2} = 0$ for $p > 0$, trivial coefficients give

$$H^*(\mathfrak{sl}_2[z], \mathfrak{sl}_2; \mathbb{C}) = \mathbb{C}, \tag{3.1}$$

concentrated in cohomological degree and polynomial-mode degree zero.

Lemma 3.2 (Hochschild–Serre, cf. [HS53]). *Let $\mathfrak{a} = \mathfrak{k} \ltimes \mathfrak{n}$ with $\mathfrak{k}$ semisimple, and let M be an $\mathfrak{a}$-module such that each cochain space $C^q(\mathfrak{a}; M)$ is a locally finite $\mathfrak{k}$-module. Then*

$$H^*(\mathfrak{a}; M) \cong H^*(\mathfrak{k}; \mathbb{C}) \otimes H^*(\mathfrak{a}, \mathfrak{k}; M).$$

This is proved as in [HS53]: one decomposes the cochains into $\mathfrak{k}$-isotypic components; the nontrivial components are contracted using the Casimir operator, and the trivial component contributes $H^*(\mathfrak{k}; \mathbb{C})$. The hypothesis holds in all cases below. For full-Hom cochains of $\mathfrak{l}$ with finite-dimensional coefficients M, the space C^q is a product of finite-dimensional $\mathfrak{sl}_2$-modules with highest weights at most $2q$ plus the largest weight of M; only finitely many isotypic types occur, although with infinite multiplicities. For graded cochains the hypothesis holds degree by degree.

Theorem 3.3 (Teleman [Tel95, Theorem 0], [Tel96, Theorem 3.1]). *Let $a_1, \dots, a_n \in \mathbb{C}$ be pairwise distinct and $m_1, \dots, m_n \geq 0$. For full-*Hom *relative cochains,*

$$H^j\Big(\mathfrak{l}, \mathfrak{sl}_2; \bigotimes_{i=1}^{n} V_{m_i}(a_i)\Big) \cong \begin{cases} \mathbb{C}, & \text{if } m_i = 2p_i \text{ for all } i \text{ and } j = \sum_i p_i, \\ 0, & \text{otherwise.} \end{cases}$$

This is the case $\Sigma = \mathbb{C}$, $h = 0$ of [Tel96, Theorem 3.1], the genus-zero statement recalled there from [Tel95, §1.4, Theorem 0]. That theorem computes the cohomology of the pair $(\mathfrak{g}[z], \mathfrak{g})$ with coefficients in $\mathsf{H} \otimes \bigotimes_k V_k(z_k)$, where the points z_k are distinct, the V_k are irreducible, and H is a

positive-energy representation at ∞ of any level $h \geq 0$ [Tel96, §1]. The cohomology is that of the standard complex $\mathrm{Hom}_{\mathfrak{g}}(\Lambda^\bullet(\mathfrak{g}[z]/\mathfrak{g}), -)$ [Tel95, (3.1.4)], which is our full-Hom complex. At level zero H is trivial. For $\mathfrak{sl}_2$ the dual Coxeter number is $c = 2$, and the ρ-shifted highest weight of V_m pairs with the positive root to give $m+1$. The alcove walls are where this pairing lies in $(h+c)\mathbb{Z} = 2\mathbb{Z}$. Hence V_m is singular if m is odd, and then the cohomology vanishes. If $m = 2p$, the shifted weight lies in the alcove $(2p, 2p+2)$, which is separated from the positive alcove $(0,2)$ by the p walls $2, 4, \dots, 2p$. So V_{2p} is regular of length p, and the small representation obtained from it by reflection [Tel96, §2] is $\mathbb{C}$, the only small representation at level zero. By [Tel96, Theorem 3.1] the cohomology is then concentrated in degree $\sum_i p_i$, where it is $\mathbb{C}^{\mathfrak{l}} = \mathbb{C}$. We do not use any multiplicative property of Theorem 3.3: the factorization of cup products that we need is proved in Lemma 6.1.

4. Injectivity: separating cycles

4.1. **The dual of A.** We describe the graded dual of A by means of the generating fields

$$X(z) := \sum_{k\geq 0} \partial^k X \frac{z^k}{k!}, \qquad Y(z) := \sum_{k\geq 0} \partial^k Y \frac{z^k}{k!}.$$

Let $V_{r,s} \subset V$ be the span of monomials with r factors $\partial^k X$ and s factors $\partial^k Y$. For a functional η on $V_{r,s}$ put

$$\Omega_\eta(z_1, \dots, z_r; y_1, \dots, y_s) := \eta\big(X(z_1)\cdots X(z_r)\, Y(y_1)\cdots Y(y_s)\big).$$

If η belongs to the graded dual, Ω_η is a polynomial, symmetric in the z_i and alternating in the y_a, and $\eta \mapsto \Omega_\eta$ identifies the graded dual of $V_{r,s}$ with the space of such polynomials; a monomial of degree d pairs with elements of spin $2r + 3s + d$. Put

$$\mathcal{O}_{r,s} := \mathbb{C}[z_1, \dots, z_r]^{S_r} \otimes \mathbb{C}[y_1, \dots, y_s]^{S_s}, \qquad F_{r,s} := \Delta(z)^2\, \Delta(y)^3 \prod_{i=1}^{r}\prod_{a=1}^{s} (z_i - y_a)^2, \tag{4.1}$$

where $\Delta(z) = \prod_{i<j}(z_i - z_j)$, and empty products are equal to one. We also write $\mathbb{C}[z, y]$ for the polynomial ring in $z_1, \dots, z_r$ and $y_1, \dots, y_s$.

Lemma 4.1. *The map $\eta \mapsto \Omega_\eta$ identifies $A^\vee_{r,s}$ with $F_{r,s}\,\mathcal{O}_{r,s}$.*

Proof. This is a reformulation of [K26, (2.54)]; we include the short argument. A functional on $V_{r,s}$ descends to $A_{r,s}$ if and only if it annihilates $I \cap V_{r,s}$, which is spanned by products of derivatives of X^2, XY, $X\partial Y$, $Y\partial Y$ with monomials. Since $X(z)^2 = e^{z\partial}(X^2)$, $X(z)Y(z) = e^{z\partial}(XY)$, $X(z)Y'(z) = e^{z\partial}(X\partial Y)$ and $Y(z)Y'(z) = e^{z\partial}(Y\partial Y)$, this happens if and only if, for all $i \neq j$ and $a \neq b$,

(a) Ω_η vanishes on $z_i = z_j$;
(b) Ω_η and $\partial_{y_a}\Omega_\eta$ vanish on $y_a = z_i$;
(c) $\partial_{y_b}\Omega_\eta$ vanishes on $y_a = y_b$.

By symmetry, (a) is equivalent to divisibility by $(z_i - z_j)^2$, and (b) is equivalent to divisibility by $(z_i - y_a)^2$. Since Ω_η is alternating, it equals $(y_a - y_b)g$ with g symmetric in y_a, y_b, and (c) says that

g vanishes on $y_a = y_b$; hence (c) is equivalent to divisibility by $(y_a - y_b)^3$. The linear forms $z_i - z_j$, $z_i - y_a$, $y_a - y_b$ are pairwise non-associate primes, so (a)–(c) together say that $F_{r,s}$ divides Ω_η, and the quotient is symmetric in the z's and in the y's. □

4.2. **Cartan currents.** Choose a Cartan element h and root vectors E, F of $\mathfrak{sl}_2$, normalized here by $[h, E] = E$ and $[h, F] = -F$, let $\mathfrak{h} = \mathbb{C}h$, and consider the subalgebra and pair

$$\mathfrak{t} := \mathfrak{h}[z] \ltimes w\,\mathfrak{sl}_2[z] \;\subset\; \mathfrak{sl}_2[z,w]/(w^2), \qquad (\mathfrak{t}, \mathfrak{h}) \;\subset\; \big(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2\big).$$

Restriction of basic cochains commutes with the differentials; dually, the inclusion of relative chains $\iota_* : C_*(\mathfrak{t}, \mathfrak{h}) \to C_*(\mathfrak{sl}_2[z,w]/(w^2), \mathfrak{sl}_2)$ is a chain map. Since $\mathcal{C}$ is the graded dual of the second complex, every cycle $\gamma \in C_*(\mathfrak{t}, \mathfrak{h})$ defines a functional $[\alpha] \mapsto \langle \alpha, \iota_* \gamma \rangle = \langle \iota^* \alpha, \gamma \rangle$ on H, which is well defined because coboundaries pair to zero with cycles.

The relative chains are $C_*(\mathfrak{t}, \mathfrak{h}) = \Lambda^\bullet(z\mathfrak{h}[z] \oplus w\,\mathfrak{sl}_2[z])_\mathfrak{h}$, the weight-zero part of the exterior algebra. Write $\xi_k := hz^k$ for $k \geq 1$. The only brackets in $\mathfrak{t}$ between the generators of this exterior algebra which do not vanish are

$$[\xi_k, wEz^m] = wEz^{m+k}, \qquad [\xi_k, wFz^m] = -wFz^{m+k}.$$

Hence the chain complex is a tensor product of three pieces: the exterior algebra on the ξ_k; the *root part* $\Lambda(wE \otimes \mathbb{C}[z]) \otimes \Lambda(wF \otimes \mathbb{C}[z])$; and the *neutral part* $\Lambda(wh \otimes \mathbb{C}[z])$, which is inert. Weight zero forces equally many factors s of each root color. Identifying wedges with alternants,

$$wEz^{m_1} \wedge \cdots \wedge wEz^{m_s} \longmapsto \det\big(u_a^{m_b}\big),$$

and similarly for F with variables $v_1, \ldots, v_s$, the root part with s factors of each color becomes $\Delta(u)\Delta(v)\,\mathcal{O}_s^{\mathrm{C}}$, where $\mathcal{O}_s^{\mathrm{C}} := \mathbb{C}[u_1, \ldots, u_s]^{S_s} \otimes \mathbb{C}[v_1, \ldots, v_s]^{S_s}$. Under this identification ξ_k acts by multiplication by

$$f_k := p_k(u) - p_k(v), \qquad p_k(u) = \textstyle\sum_a u_a^k.$$

Therefore, up to signs, the chain complex is the Koszul complex

$$\mathcal{O}_s^{\mathrm{C}} \otimes \Lambda(\xi_1, \xi_2, \ldots), \qquad d\xi_k = f_k, \tag{4.2}$$

tensored with the neutral part, on which the differential vanishes.

We now produce cycles by passing from power sums to elementary symmetric functions. Introduce a formal parameter τ, put $\mathcal{E}_u(\tau) = \prod_a (1 - u_a\tau)$, $\mathcal{E}_v(\tau) = \prod_a (1 - v_a\tau)$, and

$$L(\tau) := \log \frac{\mathcal{E}_v(\tau)}{\mathcal{E}_u(\tau)} = \sum_{k \geq 1} \frac{f_k}{k}\,\tau^k, \qquad \Xi(\tau) := \sum_{k \geq 1} \frac{\xi_k}{k}\,\tau^k,$$

$$\Theta(\tau) := \mathcal{E}_v(\tau)\,\frac{e^{-L(\tau)} - 1}{L(\tau)}\,\Xi(\tau) = \sum_{k \geq 1} \theta_k \tau^k,$$

where $(e^{-L} - 1)/L$ stands for $\sum_{n \geq 0} (-1)^{n+1} L^n/(n+1)!$. Since L has no constant term, each θ_k is a finite expression $\theta_k = -\xi_k/k + \sum_{j<k} \pi_{kj}\xi_j$ with $\pi_{kj} \in \mathcal{O}_s^{\mathrm{C}}$. The Koszul differential is $\mathcal{O}_s^{\mathrm{C}}$-linear

with $d\Xi = L$, so

$$d\Theta = \mathcal{E}_v\big(e^{-L} - 1\big) = \mathcal{E}_u - \mathcal{E}_v, \qquad \text{i.e.} \qquad d\theta_k = \begin{cases} (-1)^k\big(e_k(u) - e_k(v)\big), & k \le s, \\ 0, & k > s. \end{cases} \tag{4.3}$$

In particular $\theta_{s+1}, \theta_{s+2}, \dots$ are cycles. (For $s = 0$ one simply has $\theta_k = -\xi_k/k$.)

Pairings. Let $\beta_x(z) = \sum_{j\ge 0}(wxz^j)^\vee z^j$ for $x \in \{h, E, F\}$, and $\zeta_h(z) = \sum_{n\ge 1}(hz^n)^\vee z^n$, be the generating functions of the coordinates on $\mathfrak{t}/\mathfrak{h}$ dual to the basis $\{hz^n,\, wxz^j\}$. Restricting the fields $\mathsf{X}(z) := \sum_k \partial^k \mathsf{X}\, z^k/k! = -\sum_a b^a(z)\, \frac{d}{dz} c^a(z)$ and $\mathsf{Y}(z) = b^1(z)b^2(z)b^3(z)$, where $b^a(z) = \sum_j b^a_j z^j$ and $c^a(z) = \sum_{n\ge 1} c^a_n z^n$, kills the root components of c, and gives

$$\iota^*\mathsf{X}(z) = \kappa_X\, \beta_h(z)\, \zeta_h'(z), \qquad \iota^*\mathsf{Y}(y) = \kappa_Y\, \beta_h(y)\beta_E(y)\beta_F(y) \tag{4.4}$$

for nonzero constants κ_X, κ_Y. For $0 \le n_1 < \cdots < n_r$ and $G \in \mathbb{C}[u_1, \dots, u_s]^{S_s}$ consider the chain

$$\gamma_{n,G} := \theta_{s+1+n_1} \wedge \cdots \wedge \theta_{s+1+n_r} \wedge \big(\Delta(u)\Delta(v)\, G\big) \wedge \varpi, \qquad \varpi := wh \wedge whz \wedge \cdots \wedge whz^{r+s-1}.$$

It is a cycle, because the Koszul differential is an $\mathcal{O}_s^{\mathrm{C}}$-linear derivation of (4.2) and each θ_{s+1+n_j} is closed by (4.3). Note that closedness holds identically in u, v; the specialization $u = v$ below is only used to evaluate pairings.

Lemma 4.2. *For a nonzero constant κ independent of n and G,*

$$\big\langle \varphi\big(X(z_1)\cdots X(z_r)\, Y(y_1)\cdots Y(y_s)\big),\, \iota_*\gamma_{n,G}\big\rangle = \pm\kappa\, F_{r,s}\, s_n(z)\, G(y), \qquad s_n(z) := \frac{\det(z_i^{n_j})}{\Delta(z)}.$$

Proof. By (4.4) the restricted cochain is a product of coordinate functions of four kinds, ζ_h', β_h, β_E and β_F, which pair only with chain factors of the corresponding kinds ξ_k, whz^j, wEz^j and wFz^j. The pairing therefore factors into three determinants. (i) The neutral factor gives $\langle \prod_i \beta_h(z_i) \prod_a \beta_h(y_a), \varpi\rangle = \pm\Delta(z_1, \dots, z_r, y_1, \dots, y_s)$, a Vandermonde determinant in all $r + s$ variables. (ii) Pairing $\prod_a \beta_E(y_a)$ with an alternant $\rho(u)$ returns $\pm\rho(y)$, and similarly for F; so a root element $\Delta(u)\Delta(v)P(u, v)$ pairs to $\pm\Delta(y)^2\, \mathrm{ev}(P)$, where $\mathrm{ev} : \mathcal{O}_s^{\mathrm{C}} \to \mathbb{C}[y]^{S_s}$ is the specialization $u = v = y$. (iii) The coefficients $\pi_{kj} \in \mathcal{O}_s^{\mathrm{C}}$ of the θ's multiply the root factor, so by (ii) they enter through ev, which is a ring homomorphism. Since $\langle \Xi(\tau), \zeta_h'(z)\rangle = \sum_k \frac{\tau^k}{k} kz^{k-1} = \tau/(1 - z\tau)$, and the prefactor of Ξ in Θ specializes to $-\mathcal{E}_y(\tau)$ at $u = v = y$,

$$\mathrm{ev}\langle \theta_{s+1+n}, \zeta_h'(z)\rangle = -[\tau^{s+n}]\, \frac{\mathcal{E}_y(\tau)}{1 - z\tau} = -z^n \prod_{a=1}^{s}(z - y_a). \tag{4.5}$$

The Cartan factor is therefore $\det\big(\mathrm{ev}\langle \theta_{s+1+n_j}, \zeta_h'(z_i)\rangle\big) = \pm\prod_{i,a}(z_i - y_a)\det(z_i^{n_j})$. Multiplying (i), (ii) and (iii), and using $\Delta(z, y) = \Delta(z)\Delta(y)\prod_{i,a}(z_i - y_a)$, gives $\pm F_{r,s}\, s_n(z)\, G(y)$ up to the product of the constants $\kappa_X^r \kappa_Y^s$ and combinatorial factors. □

Proposition 4.3. *The map φ is injective. Equivalently, every $\psi_{B,p}$ is surjective.*

Proof. As n ranges over strictly increasing sequences, the Schur polynomials $s_n(z)$ form a basis of $\mathbb{C}[z_1, \dots, z_r]^{S_r}$ [Mac95, I.3], and $G(y)$ ranges over $\mathbb{C}[y_1, \dots, y_s]^{S_s}$. By Lemma 4.2, the functionals

$a \mapsto \langle \varphi(a), \iota_* \gamma_{n,G} \rangle$ on $A_{r,s}$ correspond to polynomials spanning $F_{r,s}\mathcal{O}_{r,s}$, which is all of $A^\vee_{r,s}$ by Lemma 4.1. Hence an element $a \in A_{r,s}$ with $\varphi(a) = 0$ pairs to zero with all of $A^\vee_{r,s}$, so $a = 0$. Different (r,s) lie in different (B,p), and φ preserves (B,p). □

5. Sheaves on configuration spaces of ghosts

5.1. **The finite BGG complex.** As an $\mathfrak{l}$-module, $w\,\mathfrak{sl}_2[z] \cong \mathfrak{sl}_2 \otimes \mathbb{C}[z]$ with $(xz^m)\cdot(wyz^n) = w[x,y]z^{m+n}$. Put

$$N_B := \Lambda^B\big(w\,\mathfrak{sl}_2[z]\big), \qquad S(wyz^j) = j+1.$$

The B-number B part $\mathcal{C}_B$ of $\mathcal{C}$ is the graded dual of the relative chain complex $C_*(\mathfrak{l}, \mathfrak{sl}_2; N_B)$, and Q is the transpose of its differential. Hence

$$K_{B,p} = H_p(\mathfrak{l}, \mathfrak{sl}_2; N_B). \tag{5.1}$$

By Theorem 3.1, $K_{B,p}$ is the homology of the complex C_B with terms

$$C_{B,p} = (N_B \otimes V_{2p})^{\mathfrak{sl}_2}[d_p], \qquad 0 \le p \le B; \tag{5.2}$$

there are no terms with $p > B$, since the highest weights occurring in N_B are at most $2B$.

Let $\mathcal{O}_B := \mathbb{C}[t_1, \dots, t_B]^{S_B}$, with $\deg t_i = 1$, be the coordinate ring of $\mathrm{Sym}^B \mathbb{A}^1$. Antisymmetrization identifies

$$N_B \cong \big(V_2^{\otimes B} \otimes \mathbb{C}[t_1, \dots, t_B] \otimes \mathrm{sgn}\big)^{S_B}, \qquad (xz^m)\cdot v = \sum_{i=1}^{B} t_i^m\, x^{(i)} v, \tag{5.3}$$

where $V_2 = \mathfrak{sl}_2$ is the adjoint representation, $x^{(i)}$ acts on the i-th tensor factor, and t_i^j records the mode z^j of the i-th factor. Multiplication by $\mathcal{O}_B$ commutes with the action of $\mathfrak{l}$.

Lemma 5.1. *Each $C_{B,p}$ is a finitely generated free graded $\mathcal{O}_B$-module, and the differentials of C_B are $\mathcal{O}_B$-linear. In particular every $K_{B,p}$ is a finitely generated graded $\mathcal{O}_B$-module.*

Proof. By Chevalley's theorem [Che55], $\mathbb{C}[t_1, \dots, t_B] \cong \mathcal{O}_B \otimes \mathcal{H}_B$ as graded $\mathcal{O}_B[S_B]$-modules, where $\mathcal{H}_B$ is a finite-dimensional graded S_B-stable complement of the ideal generated by invariants of positive degree. Since taking invariants under the finite group S_B and under $\mathfrak{sl}_2$ is exact and commutes with $\mathcal{O}_B \otimes -$, we get $C_{B,p} \cong \mathcal{O}_B \otimes \big(V_2^{\otimes B} \otimes \mathcal{H}_B \otimes \mathrm{sgn} \otimes V_{2p}\big)^{S_B \times \mathfrak{sl}_2}$, up to the grading shift. The differentials are induced by the $\mathfrak{l}$-action and the BGG maps, and therefore commute with $\mathcal{O}_B$. □

Geometrically, $\mathcal{K}_{B,p} := \widetilde{K_{B,p}}$ is the p-th homology sheaf of a complex of vector bundles of length B on $\mathrm{Sym}^B \mathbb{A}^1$, as announced in §1.1.

5.2. **Positions.** The derivation T_m of (2.9) is the transpose of the endomorphism of N_B induced by multiplication by z^m on $w\mathfrak{sl}_2[z]$, which by (5.3) is multiplication by the power sum $p_m(t) = \sum_i t_i^m$. Since $p_1, \dots, p_B$ generate $\mathcal{O}_B$, the $\mathcal{O}_B$-module structure of $K_{B,p}$ is the transpose of the action of the T_m on H_B^p. This is the precise sense in which the T_m measure the positions of the b-ghosts.

On the source, $T_m\,\partial^k\mathsf{X} = \frac{k!}{(k-m)!}\partial^{k-m}\mathsf{X}$ and $T_m\,\partial^k\mathsf{Y} = 3\,\frac{k!}{(k-m)!}\partial^{k-m}\mathsf{Y}$, both zero if $k < m$, as one checks from (2.10) and (2.8); on fields, $T_m\mathsf{X}(z) = z^m\mathsf{X}(z)$ and $T_m\mathsf{Y}(z) = 3z^m\mathsf{Y}(z)$. Hence T_m preserves $\varphi(A)$, and on $A^\vee_{r,s} \subset \mathbb{C}[z,y]$ the power sum $p_m(t)$ acts by multiplication by $p_m(z) + 3p_m(y)$. In other words, $\mathcal{O}_B$ acts on $A^\vee_{r,s}$ through the ring map

$$\mathcal{O}_B \longrightarrow \mathcal{O}_{r,s}, \qquad \prod_{k=1}^{B}(u - t_k) \longmapsto \prod_{i=1}^{r}(u - z_i)\prod_{a=1}^{s}(u - y_a)^3, \tag{5.4}$$

and $\psi_{B,p}$ is $\mathcal{O}_B$-linear. Finally, since $\mathcal{O}_B$ acts on N_B by $\mathfrak{l}$-module endomorphisms and the comparison of Theorem 3.1 is natural in the coefficient module, the identification $K_{B,p} \cong H_p(C_B)$ of §5.1 is $\mathcal{O}_B$-linear.

5.3. **The source sheaf.** The ring map (5.4) is the comorphism of

$$\nu_{r,s} : \mathrm{Sym}^r \mathbb{A}^1 \times \mathrm{Sym}^s \mathbb{A}^1 \longrightarrow \mathrm{Sym}^B \mathbb{A}^1, \qquad (\Pi_1(u), \Pi_3(u)) \longmapsto \Pi_1(u)\,\Pi_3(u)^3, \qquad B = r + 3s, \tag{5.5}$$

where we identify effective divisors with monic polynomials. The map $\nu_{r,s}$ is finite: each z_i and y_a is a root of the monic polynomial on the right of (5.4), whose coefficients lie in the image of $\mathcal{O}_B$, so $\mathbb{C}[z,y]$ and its subring $\mathcal{O}_{r,s}$ are finite over $\mathcal{O}_B$. On the dense open set where Π_1 and Π_3 are squarefree and coprime, they are recovered from $\Pi_1\Pi_3^3$ as the products over its simple and triple roots, respectively; so $\nu_{r,s}$ is birational onto its image, the closure of the stratum of divisors with r simple and s triple points. Its source is smooth, so $\nu_{r,s}$ is the normalization of this closure. Since multiplication by the nonzerodivisor $F_{r,s}$ identifies $A^\vee_{r,s}$ with $\mathcal{O}_{r,s}$ up to a shift of grading, we obtain

$$A^\vee_{B,p} \cong \begin{cases} \mathcal{O}_{r,s} = \Gamma\big(\nu_{r,s*}\mathcal{O}\big) & \text{if } (B,p) = (r+3s, r), \\ 0 & \text{otherwise,} \end{cases} \tag{5.6}$$

as ungraded $\mathcal{O}_B$-modules. (We make no claim about the uniqueness of the factorization $\Pi_1\Pi_3^3$ at the boundary of the stratum.)

5.4. **The top degree.**

Lemma 5.2. *Let $B \geq 1$. The top differential $C_{B,B} \to C_{B,B-1}$ vanishes, so that*

$$C_B = C_{B,B}[B] \oplus C'_B, \qquad C'_B \text{ concentrated in degrees } 0, \dots, B-1,$$

and $\psi_{B,B}$ is an isomorphism.

Proof. The V_{2B}-isotypic part of $V_2^{\otimes B}$ is the Cartan power, of multiplicity one and with trivial S_B-action. By (5.3), $C_{B,B}$ is therefore isomorphic to the module of alternating polynomials $\Delta(t)\mathcal{O}_B$, which is free of rank one. On the other side, $A^\vee_{B,B} = F_{B,0}\mathcal{O}_{B,0} = \Delta(z)^2\mathbb{C}[z]^{S_B}$ is free of rank one over

$\mathcal{O}_B$. Since $C_{B,B}$ is the top term, $K_{B,B}$ is the kernel of the top differential, and by Proposition 4.3 it surjects onto $A^{\vee}_{B,B}$; so $K_{B,B}$ has rank one, like $C_{B,B}$, over the fraction field of the domain $\mathcal{O}_B$. The image of the top differential thus has rank zero; being a submodule of a free module it is torsion-free, hence zero. Now $\psi_{B,B}$ is a surjection between free modules of rank one, and its kernel, having rank zero and lying in a free module, vanishes. □

6. Separation: factorization at disjoint supports

In this section, cochains with finite-dimensional coefficients are always the full-Hom cochains of §3. We return to the polynomial families of §5 only through the finite free BGG complexes.

6.1. **Cup products at disjoint supports.** The annihilator in $\mathfrak{l}$ of a finite-dimensional $\mathfrak{l}$-module M is an ideal of finite codimension, hence of the form $\mathfrak{sl}_2 \otimes (f)$ for a monic polynomial f; we call the set of roots of f the *support* of M.

Lemma 6.1. *Let M_1 and M_2 be finite-dimensional $\mathfrak{l}$-modules with disjoint supports. Then the cup product*

$$C^*(\mathfrak{l}, \mathfrak{sl}_2; M_1) \otimes C^*(\mathfrak{l}, \mathfrak{sl}_2; M_2) \longrightarrow C^*(\mathfrak{l}, \mathfrak{sl}_2; M_1 \otimes M_2)$$

is a quasi-isomorphism.

Proof. Step 1: explicit cocycles. Use the standard triple E, h, F with $[h, E] = 2E$, and let $x_j \in (\mathfrak{l}/\mathfrak{sl}_2)^{\vee}$ be the coordinate dual to Fz^j, $j \geq 1$, with respect to the basis of modes; it has weight 2. Let $\omega_p := x_1 \wedge \cdots \wedge x_p$. The relative differential of x_j is a combination of the products $(hz^i)^{\vee} \wedge x_{j-i}$, $1 \leq i < j$, and each such term repeats a factor already present in ω_p; hence ω_p is closed. It is a highest-weight vector of weight $2p$, so it generates a copy of V_{2p}. Contracting this copy with the coefficient module $V_{2p}(0)$, on which $z\,\mathfrak{sl}_2[z]$ acts by zero, gives an invariant cocycle $\alpha_{p,0} \in C^p(\mathfrak{l}, \mathfrak{sl}_2; V_{2p}(0))$. Transporting it by the automorphism $z \mapsto z - a$ of $\mathfrak{l}$, which fixes $\mathfrak{sl}_2$, gives a cocycle $\alpha_{p,a}$ with coefficients in $V_{2p}(a)$.

Step 2: products at distinct points are nonzero. Let $a_1, \ldots, a_n$ be distinct and $M = \bigotimes_i V_{2p_i}(a_i)$. Restrict cochains to the pair $(\mathfrak{h}[z], \mathfrak{h})$ and decompose $M = \bigoplus_\mu M_\mu$ according to the tuples $\mu = (\mu_i)$ of weights of the factors. Each M_μ is preserved by $z\mathfrak{h}[z]$, which acts through the character $\chi_\mu(hf) = \sum_i \mu_i f(a_i)$. Since $z\mathfrak{h}[z]$ is abelian, the restricted complex is $\bigoplus_{\sum \mu_i = 0} \mathrm{Hom}(\Lambda^\bullet z\mathfrak{h}[z], M_\mu)$ with differential $\alpha \mapsto \chi_\mu \wedge \alpha$; in particular the summand with $\mu = 0$ is a direct summand with zero differential. Under restriction and projection to this summand, $\alpha_{p,a}$ becomes a nonzero multiple of $\ell_{a,1} \wedge \cdots \wedge \ell_{a,p}$, where $\ell_{a,j}(hf) := f^{(j)}(a)/j!$. Indeed, only the weight-zero vector of the copy of V_{2p} generated by ω_p pairs with the zero-weight line of V_{2p}, and this vector is proportional to $F^p\omega_p$; the component of $F^p\omega_p$ involving only Cartan modes arises by lowering each factor x_j exactly once, and the $p!$ such terms contribute with equal nonzero coefficients. Consequently the product $\prod_i \alpha_{p_i,a_i}$ projects to a nonzero multiple of $\bigwedge_{i,j} \ell_{a_i,j}$, $1 \leq j \leq p_i$. By Hermite interpolation the functionals $\ell_{a_i,j}$, $0 \leq j \leq p_i$, are linearly independent on $\mathbb{C}[z]$; those with $j \geq 1$ vanish on constants

and are therefore independent on $z\mathbb{C}[z]$, so their wedge product is nonzero. Hence the class of $\prod_i \alpha_{p_i,a_i}$ is nonzero.

Step 3: simple modules. The quotient $\mathfrak{l}/\mathfrak{sl}_2 \otimes (f)$ is a product of jet algebras $\mathfrak{sl}_2 \otimes \mathbb{C}[z]/(z-a)^k$ over the roots a of f, so a simple finite-dimensional $\mathfrak{l}$-module is a tensor product, over the points of its support, of simple modules of these jet algebras. On a simple module M of a jet algebra the radical $\mathfrak{r} = \mathfrak{sl}_2 \otimes (z-a)\mathbb{C}[z]/(z-a)^k$ acts trivially. Indeed, since $\mathfrak{r}$ is nilpotent, M is the direct sum of its generalized weight spaces for finitely many weights $\lambda : \mathfrak{r} \to \mathbb{C}$, which vanish on $[\mathfrak{r},\mathfrak{r}]$ [Jac62, Chapter II]; the connected group SL_2 permutes this finite set and therefore fixes each weight, so $\lambda([\mathfrak{sl}_2,\mathfrak{r}]) = 0$; as $[\mathfrak{sl}_2,\mathfrak{r}] = \mathfrak{r}$, all weights vanish, $\mathfrak{r}$ acts nilpotently, and by Engel's theorem the joint kernel of $\mathfrak{r}$ is a nonzero submodule, hence all of M. Thus simple finite-dimensional $\mathfrak{l}$-modules are of the form $\bigotimes_i V_{m_i}(a_i)$ with distinct a_i. Now let M_1 and M_2 be simple with disjoint supports. If some m_i is odd, then by Theorem 3.3 the target and one of the two factors of the source have vanishing cohomology. Otherwise, by Theorem 3.3 the cohomologies of M_1, M_2 and $M_1 \otimes M_2$ are one-dimensional, by Step 2 they are spanned by the products of the cocycles α_{p_i,a_i}, and the cup product maps the generator of the source, which is the class of a product, to the generator of the target. In both cases the cup product is an isomorphism on cohomology; by the Künneth formula over $\mathbb{C}$, it is a quasi-isomorphism.

Step 4: dévissage. Short exact sequences of coefficient modules split over the constant subalgebra $\mathfrak{sl}_2$, so $C^*(\mathfrak{l},\mathfrak{sl}_2;-)$ takes them to short exact sequences of complexes; the same holds for $-\otimes M_2$ and $-\otimes_{\mathbb{C}} C^*(\mathfrak{l},\mathfrak{sl}_2;M_2)$. Since the cup product is natural, induction on the lengths of composition series of M_1 and M_2, together with the five lemma applied to the long exact cohomology sequences, extends Step 3 to arbitrary M_1 and M_2 with disjoint supports. □

6.2. **Coprime splitting.** The comparison for polynomial families rests on the following version of the Chinese remainder theorem, which treats simple roots and weighted clusters of roots simultaneously and holds over nonreduced rings.

Lemma 6.2. *Let R be a commutative ring and let $f_1, f_2 \in R[u]$ be monic of positive degrees d_1, d_2 with invertible resultant. Let $m_1,\dots,m_n$ be positive integers with $\sum_j m_j = d_1 + d_2$. Let $\mathcal{R}$ be the R-algebra generated by $x_1,\dots,x_n$ subject to the coefficient identities of*

$$\prod_{j=1}^{n}(u-x_j)^{m_j} = f_1(u)f_2(u),$$

and for $J \subseteq \{1,\dots,n\}$ let $\mathcal{R}_J$ be the R-algebra defined by $\prod_{j\in J}(u-x_j)^{m_j} = f_1(u)$ and $\prod_{j\notin J}(u-x_j)^{m_j} = f_2(u)$. Then the natural map

$$\mathcal{R} \longrightarrow \prod_{J:\,\sum_{j\in J} m_j = d_1} \mathcal{R}_J$$

is an isomorphism.

Proof. Choose $\alpha, \beta \in R[u]$ with $\alpha f_1 + \beta f_2 = 1$. In $\mathcal{R}$ we have $(f_1 f_2)(x_j) = 0$, so the elements $e_j := \beta(x_j) f_2(x_j)$ are idempotents, since $e_j(1-e_j) = (\alpha\beta f_1 f_2)(x_j) = 0$. They split $\mathcal{R}$ into components

indexed by the subsets J on which $e_j = 1$. In the component indexed by J, for $j \in J$ the element $f_2(x_j)$ is invertible and hence $f_1(x_j) = 0$, while for $j \notin J$ the element $f_1(x_j)$ is invertible and $f_2(x_j) = 0$. Put $g_1 = \prod_{j\in J}(u - x_j)^{m_j}$ and $g_2 = \prod_{j\notin J}(u - x_j)^{m_j}$. For $j \notin J$, the identity $f_1(u) - f_1(x_j) = (u - x_j)q_j(u)$ shows that $u - x_j$ is invertible modulo f_1, with inverse $-q_j(u)/f_1(x_j)$; hence g_2 is invertible modulo f_1, and $g_1g_2 = f_1f_2$ implies $f_1 \mid g_1$. Symmetrically $f_2 \mid g_2$. On a nonzero component, division by a monic polynomial gives $\deg g_i \geq d_i$; as $\deg g_1 + \deg g_2 = d_1 + d_2$, both are equalities, so $g_i = f_i$, and components with $\sum_{j\in J} m_j \neq d_1$ vanish. Conversely, the identities $g_i = f_i$ force the prescribed values of the idempotents. This identifies the components with the algebras $\mathcal{R}_J$. □

6.3. **Factorization.** Fix $B = B_1 + B_2$ with $B_1, B_2 \geq 1$. Let f_i be the universal monic polynomial of degree B_i over $\mathcal{O}_{B_i}$, and put

$$\mathcal{O}^{\circ}_{B_1,B_2} := (\mathcal{O}_{B_1} \otimes_{\mathbb{C}} \mathcal{O}_{B_2})\big[\mathrm{Res}(f_1, f_2)^{-1}\big], \qquad U_{B_1,B_2} := \operatorname{Spec} \mathcal{O}^{\circ}_{B_1,B_2}, \tag{6.1}$$

the space of pairs of divisors with disjoint supports. The map $U_{B_1,B_2} \to \operatorname{Sym}^B \mathbb{A}^1$, $(f_1, f_2) \mapsto f_1f_2$, is étale: its tangent map $(h_1, h_2) \mapsto h_1f_2 + f_1h_2$, with $\deg h_i < B_i$, is an isomorphism when f_1 and f_2 are coprime. In particular $\mathcal{O}^{\circ}_{B_1,B_2}$ is flat over $\mathcal{O}_B$.

Applying Lemma 6.2 with all $m_j = 1$ to $\mathbb{C}[t_1, \ldots, t_B] \otimes_{\mathcal{O}_B} \mathcal{O}^{\circ}_{B_1,B_2}$ decomposes it into the product, over the subsets of B_1 labels, of the algebras of ordered roots of f_1 and of f_2. The group S_B permutes these factors transitively, with stabilizer $S_{B_1} \times S_{B_2}$. Since finite-group invariants commute with base change (averaging is a projection), taking sign-twisted invariants in (5.3) gives an isomorphism of $\mathfrak{l}$-modules

$$N_B \otimes_{\mathcal{O}_B} \mathcal{O}^{\circ}_{B_1,B_2} \cong \big(N_{B_1} \otimes_{\mathbb{C}} N_{B_2}\big)\big[\mathrm{Res}^{-1}\big]. \tag{6.2}$$

At a closed point (f_1, f_2) of U_{B_1,B_2}, the fiber of N_{B_i} is annihilated by $\mathfrak{sl}_2 \otimes (f_i)$, because $f_i(t_k) = 0$ in the corresponding algebra of ordered roots; so the two factors of (6.2) have disjoint supports in every closed fiber.

Proposition 6.3. *On U_{B_1,B_2} there are isomorphisms of $\mathcal{O}^{\circ}_{B_1,B_2}$-modules*

$$K_{B,p} \otimes_{\mathcal{O}_B} \mathcal{O}^{\circ}_{B_1,B_2} \cong \bigoplus_{p_1+p_2=p} \big(K_{B_1,p_1} \otimes_{\mathbb{C}} K_{B_2,p_2}\big)\big[\mathrm{Res}^{-1}\big], \tag{6.3}$$

$$A^{\vee}_{B,p} \otimes_{\mathcal{O}_B} \mathcal{O}^{\circ}_{B_1,B_2} \cong \bigoplus_{p_1+p_2=p} \big(A^{\vee}_{B_1,p_1} \otimes_{\mathbb{C}} A^{\vee}_{B_2,p_2}\big)\big[\mathrm{Res}^{-1}\big]. \tag{6.4}$$

We only use these as abstract isomorphisms of modules; their compatibility with ψ plays no role.

Proof. For (6.3) we first construct a comparison of perfect complexes. The augmented complex $P_\bullet \otimes P_\bullet \to \mathbb{C}$, with the diagonal action of $\mathfrak{l}$, is exact by the Künneth formula and split over $\mathfrak{sl}_2$, so relative projectivity of $P_\bullet$ provides a diagonal $P_\bullet \to P_\bullet \otimes P_\bullet$ lifting the identity of $\mathbb{C}$. On each inducing space V_{2p} it involves only finitely many terms. For $\mathfrak{l}$-modules M_1, M_2 it induces a cup product

$\mathrm{Hom}_{\mathfrak{l}}(P_\bullet, M_1) \otimes \mathrm{Hom}_{\mathfrak{l}}(P_\bullet, M_2) \to \mathrm{Hom}_{\mathfrak{l}}(P_\bullet, M_1 \otimes M_2)$, defined over any coefficient ring and compatible with base change; comparing with the standard relative Chevalley–Eilenberg resolution, which has the usual diagonal, identifies it up to homotopy with the cup product of Lemma 6.1. Apply this to the $\mathcal{O}$-linear duals of the finite free coefficient modules in (6.2), and dualize the resulting bounded complexes of finite free $\mathcal{O}^\circ_{B_1,B_2}$-modules. Since $\mathrm{Hom}_{\mathfrak{l}}(P_p, N^\vee) = \mathrm{Hom}_{\mathfrak{sl}_2}(V_{2p}, N^\vee)$ is the dual of $(N \otimes V_{2p})^{\mathfrak{sl}_2}$, this produces a map of complexes

$$C_B \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2} \longrightarrow \left(C_{B_1} \otimes_{\mathbb{C}} C_{B_2}\right)\left[\mathrm{Res}^{-1}\right]. \tag{6.5}$$

At a closed point, the fiber of (6.5) is the dual of the cup product for the finite-dimensional modules $M_i = (N_{B_i} \otimes_{\mathcal{O}_{B_i}} \mathbb{C}_{f_i})^\vee$, where $\mathbb{C}_{f_i}$ is the residue field of $\mathcal{O}_{B_i}$ at the point f_i; these have disjoint supports, and by Theorem 3.1 and Lemma 6.1, it is a quasi-isomorphism. The cone of (6.5) is a bounded complex of finite free modules whose fibers at all closed points are acyclic. Such a complex is acyclic: its homology is supported on a closed set, which, if nonempty, contains a closed point $\mathfrak{m}$, and after localizing at $\mathfrak{m}$ and cancelling isomorphisms the complex becomes a nonzero minimal complex, whose fiber at $\mathfrak{m}$ has zero differential and is nonzero. Hence (6.5) is a quasi-isomorphism. Taking homology, using flatness on the left and the Künneth formula over $\mathbb{C}$ followed by localization on the right, gives (6.3). Note that no infinite Chevalley–Eilenberg complex has been dualized over $\mathcal{O}^\circ_{B_1,B_2}$.

For (6.4), let $(B,p) = (r+3s, r)$, since otherwise both sides vanish. By (5.4), the algebra $\mathbb{C}[z,y] \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2}$ is defined by the identity

$$f_1(u)f_2(u) = \prod_{i=1}^{r}(u - z_i)\prod_{a=1}^{s}(u - y_a)^3.$$

Apply Lemma 6.2 with weight one on the z_i and weight three on the y_a, and take $S_r \times S_s$-invariants. The orbits of components are indexed by $r_1 + r_2 = r$ and $s_1 + s_2 = s$ with $r_i + 3s_i = B_i$, and the stabilizer of a component permutes each of its four lists of roots separately. By the same averaging argument,

$$\mathcal{O}_{r,s} \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2} \cong \bigoplus_{\substack{r_1+r_2=r,\ s_1+s_2=s \\ r_i+3s_i=B_i}} \left(\mathcal{O}_{r_1,s_1} \otimes_{\mathbb{C}} \mathcal{O}_{r_2,s_2}\right)\left[\mathrm{Res}^{-1}\right].$$

By (5.6) this is (6.4). □

7. Collision: depth along the small diagonal

7.1. **The small diagonal.** For $B \geq 1$, let e_i denote the elementary symmetric functions of $t_1, \ldots, t_B$, and let $\mathfrak{p}_B \subset \mathcal{O}_B$ be the ideal

$$\mathfrak{p}_B := \left(e_i - \tbinom{B}{i}(e_1/B)^i \,:\, 2 \leq i \leq B\right). \tag{7.1}$$

Its zero set is the small diagonal Z_B of divisors of the form $B \cdot a$, i.e. of polynomials $(u-a)^B$. Its generators are coordinates on $\mathcal{O}_B = \mathbb{C}[e_1, \ldots, e_B]$ after a triangular change of variables, so they form a regular sequence of length $c = B - 1$, and $\mathfrak{p}_B$ is prime. (For $B = 1$, $\mathfrak{p}_1 = 0$.)

Depth. For a finitely generated module M over a Noetherian ring R and an ideal I, $\operatorname{depth}_I M$ is the supremum of the lengths of M-regular sequences in I, with $\operatorname{depth}_I M = \infty$ if $IM = M$ [Stacks, Definition 10.72.1, Tag 00LI]. We use: (D1) the depth lemma for short exact sequences [Stacks, Section 47.11], [BH93, §1.2]; (D2) $\operatorname{depth}_I M = \inf\{\operatorname{depth} M_{\mathfrak{q}} : \mathfrak{q} \supseteq I\}$ [BH93, §1.2]; (D3) $\operatorname{depth}_I M = \operatorname{depth}_{\sqrt{I}} M$, which follows from (D2); and (D4) for a finite map of Noetherian rings $R \to R'$ and a finitely generated R'-module M, $\operatorname{depth}_I M = \operatorname{depth}_{IR'} M$. To prove (D4), apply [Stacks, Lemma 10.72.11, Tag 0AUK], which concerns a local base ring, to the finite maps $R_{\mathfrak{q}} \to R'_{\mathfrak{q}}$ for primes $\mathfrak{q} \supseteq I$, whose maximal ideals are the primes of R' lying over $\mathfrak{q}$, and combine with (D2) for R and for R'.

7.2. **A variant of the acyclicity lemma.** The following is a variant of the acyclicity lemma of Peskine and Szpiro [PS73, Lemme 1.8] (see also [Stacks, Tag 00N0] and [BE73]).

Lemma 7.1. *Let R be Noetherian, let $I \subset R$ be generated by an R-regular sequence of length c, and let $C_c \to \cdots \to C_1 \to C_0$ be a complex of finitely generated free R-modules. Suppose that for $2 \le p \le c$ there are surjections $H_p(C) \twoheadrightarrow J_p$ whose kernels M_p are supported in $V(I)$, and that*

$$\operatorname{depth}_I J_p \ge p \qquad (2 \le p \le c).$$

Then $M_p = 0$ for $2 \le p \le c$.

Proof. Write $\overline{C}_p := \operatorname{coker}(C_{p+1} \to C_p)$ and $\mathcal{I}_p := \operatorname{im}(C_p \to C_{p-1})$. We show by descending induction on p, from $p = c$ to $p = 2$, that $\operatorname{depth}_I \overline{C}_p \ge p - 1$ and $M_p = 0$. Initially $\overline{C}_c = C_c$ is free, so $\operatorname{depth}_I \overline{C}_c \ge c$. Suppose $\operatorname{depth}_I \overline{C}_p \ge p - 1 \ge 1$. Since $M_p \subseteq H_p(C) \subseteq \overline{C}_p$ is finitely generated and supported in $V(I)$, some power I^N annihilates it. Either I contains a nonzerodivisor x on $\overline{C}_p$, and then x^N kills M_p and acts injectively on $\overline{C}_p$; or $I\overline{C}_p = \overline{C}_p$, and then $\overline{C}_p$ vanishes at every prime containing I. In both cases $M_p = 0$, so $H_p(C) = J_p$. The exact sequences

$$0 \to H_p(C) \to \overline{C}_p \to \mathcal{I}_p \to 0, \qquad 0 \to \mathcal{I}_p \to C_{p-1} \to \overline{C}_{p-1} \to 0$$

and (D1) give $\operatorname{depth}_I \mathcal{I}_p \ge \min(p-1, p-1)$ and $\operatorname{depth}_I \overline{C}_{p-1} \ge \min(c, p-2) = p - 2$, which completes the induction. □

Lemma 7.2. *For $0 < p < B$ we have $\operatorname{depth}_{\mathfrak{p}_B} A^{\vee}_{B,p} \ge p$.*

Proof. If $A^{\vee}_{B,p} = 0$ there is nothing to prove. Otherwise $(B, p) = (r + 3s, r)$ with $s \ge 1$, and by (5.6) $A^{\vee}_{B,p} \cong \mathcal{O}_{r,s}$ as an $\mathcal{O}_B$-module. Over a point of Z_B, the condition $\Pi_1(u)\,\Pi_3(u)^3 = (u - a_0)^B$ forces $\Pi_1 = (u - a_0)^r$ and $\Pi_3 = (u - a_0)^s$; so the radical of $\mathfrak{p}_B \mathcal{O}_{r,s}$ is the ideal of the line

$$\left\{\left((u - a_0)^r, (u - a_0)^s\right) : a_0 \in \mathbb{A}^1\right\} \subset \operatorname{Sym}^r \mathbb{A}^1 \times \operatorname{Sym}^s \mathbb{A}^1 \cong \mathbb{A}^{r+s}.$$

This line is the graph of a polynomial map over the coordinate $a_0 = e_1(y)/s$, so its ideal is generated by $r + s - 1$ of the coordinates of a polynomial coordinate system on $\mathbb{A}^{r+s}$, which form a regular sequence. By (D4) and (D3),

$$\operatorname{depth}_{\mathfrak{p}_B} A^{\vee}_{B,p} = \operatorname{depth}_{\mathfrak{p}_B \mathcal{O}_{r,s}} \mathcal{O}_{r,s} = \operatorname{depth}_{\sqrt{\mathfrak{p}_B \mathcal{O}_{r,s}}} \mathcal{O}_{r,s} \ge r + s - 1 \ge r = p.$$ □

Remark 7.3. This explains why the top term had to be split off in Lemma 5.2. For the pure-X sector $s = 0$ one has $p = r = B$, but only $\operatorname{depth}_{\mathfrak{p}_B} A^{\vee}_{B,B} = B - 1$. Removing this sector shortens the free complex to length $B - 1$, which is exactly the length of the regular sequence generating $\mathfrak{p}_B$, and leaves precisely the depth bounds needed in Lemma 7.1.

8. Deformation: the two lowest degrees

8.1. **Two deformations of the square-zero algebra.** For $\ell \in \{0, 1\}$ define odd derivations of the cochain algebra by

$$\delta_\ell b^a_j = 0, \qquad \delta_\ell c^a_n = \frac{1}{2} \sum_{\substack{i,k \geq 0 \\ i+k=n-\ell}} \epsilon_{abc}\, b^b_i b^c_k, \tag{8.1}$$

with empty sums equal to zero, and put $D_\ell := Q + \delta_\ell$. Each δ_ℓ raises B by 2 and lowers p by 1; δ_1 raises the spin by 1 and δ_0 by 2. One checks directly that $[\delta_1, \partial] = \delta_0$ and $[\delta_0, \partial] = 0$.

Lemma 8.1. *(i) On $\mathcal{C}$ we have $\delta_\ell^2 = 0$ and $Q\delta_\ell + \delta_\ell Q = 0$.*
(ii) $(\mathcal{C}, D_1)$ computes $H^*(\mathfrak{sl}_2[z,w]/(w^2 - z), \mathfrak{sl}_2) \cong H^*(\mathfrak{sl}_2[u], \mathfrak{sl}_2)$, via $z = u^2$ and $w = u$. The u-degree is $\mathrm{wt}_1 := 2S - B$, and the cohomology is $\mathbb{C}$, in $\mathrm{wt}_1 = 0$.
(iii) $(\mathcal{C}, D_0)$ computes $H^*(\mathfrak{sl}_2[z,w]/(w^2 - 1), \mathfrak{sl}_2) \cong H^*(\mathfrak{sl}_2[z] \oplus \mathfrak{sl}_2[z], \mathfrak{sl}_2^{\mathrm{diag}})$, via $w \mapsto \pm 1$. The z-degree is $\mathrm{wt}_0 := S - B$, and the cohomology is $\mathbb{C} \oplus \mathbb{C}\,\mathsf{Y}$, in $\mathrm{wt}_0 = 0$.

Proof. The algebra $\mathbb{C}[z,w]/(w^2 - z^\ell)$ has the same basis of modes z^n, wz^j as $\mathbb{C}[z,w]/(w^2)$, so $\mathcal{C}$ is also the space of basic polynomial-mode cochains of $\mathfrak{sl}_2[z,w]/(w^2 - z^\ell)$ relative to $\mathfrak{sl}_2 \otimes 1$. The only new brackets are $[wxz^i, wyz^j] = [x,y]z^{i+j+\ell}$, whose dual is the term (8.1); hence D_ℓ is the Chevalley–Eilenberg differential of $\mathfrak{sl}_2[z,w]/(w^2 - z^\ell)$ on basic cochains, in the conventions of (2.5). Since $D_\ell^2 = 0$ there and Q, δ_ℓ have B-degrees 0 and 2, (i) follows by separating B-degrees. (For $\ell = 0$ this relies on basicness: on the full exterior algebra one finds $D_0^2 = \frac{1}{2}\sum_a \epsilon_{abc}\, b^b_0 b^c_0\, L_a$, which vanishes on invariants.)

For $\ell = 1$, the isomorphism $\mathbb{C}[z,w]/(w^2 - z) \cong \mathbb{C}[u]$ sends $z^n \mapsto u^{2n}$ and $wz^j \mapsto u^{2j+1}$, so the u-degree of c^a_n and b^a_j is $2n = 2S - B$ and $2j + 1 = 2S - B$, respectively. The cohomology is given by (3.1) in the variable u.

For $\ell = 0$, $\mathbb{C}[z,w]/(w^2 - 1) \cong \mathbb{C}[z] \times \mathbb{C}[z]$ via $w \mapsto (1, -1)$, the constants $\mathfrak{sl}_2 \otimes 1$ become the diagonal, and the z-degree of c^a_n and b^a_j is $n = S - B$ and $j = S - B$. By Lemma 3.2 applied to $\mathfrak{sl}_2[z] = \mathfrak{sl}_2 \ltimes z\,\mathfrak{sl}_2[z]$ and by (3.1), $H^*(\mathfrak{sl}_2[z]; \mathbb{C}) = H^*(\mathfrak{sl}_2; \mathbb{C})$, in z-degree zero, and by the Künneth formula $H^*(\mathfrak{sl}_2[z] \oplus \mathfrak{sl}_2[z]; \mathbb{C}) = H^*(\mathfrak{sl}_2; \mathbb{C})^{\otimes 2}$, again in z-degree zero. The homomorphism $(f, g) \mapsto (f(0), f(0))$ is a retraction onto $\mathfrak{sl}_2^{\mathrm{diag}}$, so $\mathfrak{sl}_2[z] \oplus \mathfrak{sl}_2[z] = \mathfrak{sl}_2^{\mathrm{diag}} \ltimes \mathfrak{n}$, and Lemma 3.2, which respects the z-degree, shows that the relative cohomology is two-dimensional and concentrated in z-degree zero. In $\mathrm{wt}_0 = 0$ the basic cochains are $\Lambda(b^1_0, b^2_0, b^3_0)^{\mathfrak{sl}_2} = \mathbb{C} \oplus \mathbb{C}\,\mathsf{Y}$, with zero differential. ∎

8.2. **The deformations on A.** Write $X_j = \partial^j X$, $Y_j = \partial^j Y$, and similarly $\mathsf{X}_j = \partial^j \mathsf{X}$, $\mathsf{Y}_j = \partial^j \mathsf{Y}$. With $b^a(\tau) = \sum_j b^a_j \tau^j$, $c^a(\tau) = \sum_{n\geq 1} c^a_n \tau^n$ and $P_a(\tau) := \frac{1}{2}\epsilon_{abc} b^b(\tau) b^c(\tau)$, the definition (8.1) reads $\delta_1 c^a(\tau) = \tau P_a(\tau)$ and $\delta_0 c^a(\tau) = P_a(\tau) - P_a(0)$. Since $\sum_j \mathsf{X}_j \tau^j/j! = e^{\tau\partial}\mathsf{X} = -\sum_a b^a(\tau)\frac{d}{d\tau}c^a(\tau)$ and $e^{\tau\partial}\mathsf{Y} = b^1(\tau)b^2(\tau)b^3(\tau)$, and since $\sum_a b^a P_a = 3\, b^1 b^2 b^3$ and $\sum_a b^a \frac{d}{d\tau} P_a = 2\,\frac{d}{d\tau}(b^1 b^2 b^3)$, we obtain

$$\delta_1 \mathsf{X}_j = (2j+3)\,\mathsf{Y}_j, \qquad \delta_0 \mathsf{X}_j = 2\,\mathsf{Y}_{j+1}, \qquad \delta_1 \mathsf{Y}_j = \delta_0 \mathsf{Y}_j = 0. \tag{8.2}$$

Since the δ_ℓ are derivations anticommuting with Q, they act on H, and by (8.2) they preserve $\varphi(A)$. By Proposition 4.3 they are transported to derivations of A, determined by $\delta_1 X_j = (2j+3)Y_j$, $\delta_0 X_j = 2Y_{j+1}$ and $\delta_\ell Y_j = 0$.

Lemma 8.2. *Let h_1, h_0 be the odd derivations of V with*

$$h_1 X_j = h_0 X_j = 0, \qquad h_1 Y_j = X_j, \qquad h_0 Y_j = \tfrac{j}{2}\, X_{j-1} \quad (h_0 Y_0 = 0).$$

They preserve I, hence descend to A, and on A

$$\delta_1 h_1 + h_1 \delta_1 = \mathrm{wt}_1 = 2S - B, \qquad \delta_0 h_0 + h_0 \delta_0 = \mathrm{wt}_0 = S - B. \tag{8.3}$$

Proof. The derivation h_1 commutes with ∂ and sends the four generators X^2, XY, $X\partial Y$, $Y\partial Y$ of I to 0, X^2, $\frac{1}{2}\partial(X^2)$ and $2X\partial Y - \partial(XY)$, respectively; so it preserves the differential ideal I. The derivation h_0 sends the same generators to 0, 0, $\frac{1}{2}X^2$ and $-\frac{1}{2}XY$, and satisfies $[h_0, \partial] = \frac{1}{2}h_1$. Hence $h_0(\partial^k \rho) = \partial h_0(\partial^{k-1}\rho) + \frac{1}{2}h_1(\partial^{k-1}\rho) \in I$ for each generator ρ by induction on k, and h_0 preserves I. Both sides of each identity in (8.3) are even derivations, so it suffices to check them on generators, where they follow from (2.1) and the formulas for δ_ℓ: for instance $(\delta_1 h_1 + h_1\delta_1)X_j = (2j+3)X_j$ and $(\delta_0 h_0 + h_0\delta_0)Y_j = jY_j$. □

8.3. **The lowest two degrees.** Regard A as a subspace of H via φ, and put $\overline{H}^p_B := H^p_B / \varphi(A^p_B)$. No algebra structure on $\overline{H}$ is needed.

Proposition 8.3. *If $\overline{H}^p_b = 0$ for all $b < B$ and all p, then $\overline{H}^0_B = \overline{H}^1_B = 0$.*

Proof. The parabola. Fix $m > 0$ and consider the subcomplex of $(\mathcal{C}, D_1)$ with $\mathrm{wt}_1 = m$. It is finite-dimensional, since every generator has positive wt_1, and it is acyclic by Lemma 8.1(ii). As $\mathrm{wt}_1 = 2S - B$, the parity ε of B is fixed. Filter by B, i.e. by $(B-\varepsilon)/2$. The resulting spectral sequence has $E_1 = H$ (at $\mathrm{wt}_1 = m$), first differential $d_1 = \delta_1$, higher differentials

$$d_r : (B, p, S) \longmapsto (B + 2r,\ p + 1 - 2r,\ S + r),$$

and converges to zero. By (8.2), $0 \to \varphi(A) \to H \to \overline{H} \to 0$ is a short exact sequence of complexes for d_1, and $\varphi(A)$ is d_1-acyclic at $\mathrm{wt}_1 = m > 0$ by Lemma 8.2. Hence $E_2 = H(\overline{H}, \delta_1)$, and by hypothesis E_2 vanishes in all B-numbers below B.

At $(B, 0)$ the incoming d_1 starts at $\overline{H}^1_{B-2} = 0$ and the outgoing d_1 lands in $p = -1$, so $E_2 = \overline{H}^0_B$. For $r \geq 2$, outgoing differentials land in negative p and incoming ones start at B-numbers below B. So $\overline{H}^0_B$ survives to $E_\infty = 0$. Since this holds for every $m > 0$, and $\mathrm{wt}_1 = 0$ contains only the constants, $\overline{H}^0_B = 0$.

Put $\mathcal{V}_S := (\overline{H}^1_B)_S$ and $\mathcal{W}_S := (\overline{H}^0_{B+2})_S$. At the positions $(B,1,S)$ and $(B+2,0,S+1)$, which have the same wt_1, the E_2-terms are the kernel and the cokernel of $\delta_1 : \mathcal{V}_S \to \mathcal{W}_{S+1}$. For $r \geq 2$, all outgoing differentials land in negative p, and all incoming ones start at B-numbers at most $B-2$. Both entries therefore survive to $E_\infty = 0$, and

$$\delta_1 : \mathcal{V}_S \xrightarrow{\sim} \mathcal{W}_{S+1} \qquad \text{for all } S. \tag{8.4}$$

(When $2S - B \leq 0$ both sides vanish.) Note that no assumption about B-number $B+2$ has been made.

The pair of lines. Now fix $m > 0$ and use D_0 in $\mathrm{wt}_0 = S - B = m$. This subcomplex is again finite-dimensional, because the only generators with $\mathrm{wt}_0 = 0$ are the three odd elements b^a_0, and it is acyclic by Lemma 8.1(iii). Unlike wt_1, the weight wt_0 does not fix the parity of B; but D_0 preserves it, since Q preserves B and δ_0 raises it by 2. We therefore split the subcomplex according to the parity ε of B and filter each part by $(B-\varepsilon)/2$. This gives a spectral sequence with $d_1 = \delta_0$ and

$$d_r : (B,p,S) \longmapsto (B+2r,\ p+1-2r,\ S+2r),$$

and Lemma 8.2 again identifies E_2 with $H(\overline{H},\delta_0)$. The argument above gives

$$\delta_0 : \mathcal{V}_S \xrightarrow{\sim} \mathcal{W}_{S+2} \qquad \text{for all } S \tag{8.5}$$

in $\mathrm{wt}_0 > 0$. In $\mathrm{wt}_0 = 0$ the whole complex is $\mathbb{C} \oplus \mathbb{C}\,Y \subset \varphi(A)$ by Lemma 8.1(iii), so $\overline{H}$ vanishes there, and there are no cochains with $\mathrm{wt}_0 < 0$. Thus (8.5) holds for all S.

Conclusion. Suppose $\overline{H}^1_B \neq 0$, and let S be the smallest spin with $\mathcal{V}_S \neq 0$. By (8.4), $\mathcal{W}_{S+1} \cong \mathcal{V}_S \neq 0$. By (8.5), $\mathcal{W}_{S+1} \cong \mathcal{V}_{S-1} = 0$. This contradiction shows $\overline{H}^1_B = 0$. All filtrations used are finite in each fixed wt_ℓ, so no convergence issues arise. □

Remark 8.4. Either deformation alone would allow a pair of excess classes in $(B,1)$ and $(B+2,0)$, cancelling each other in the spectral sequence. What rules them out is that the parabola and the pair of lines demand different spin shifts, $+1$ and $+2$, for the same pair, which is impossible for a grading bounded below. This replaces any Euler characteristic or residue computation in low degrees.

9. Proof of the main theorem

Proof of Theorem 1.1. By Proposition 4.3, φ is injective and every $\psi_{B,p}$ is surjective. We show that $\psi_{B,p}$ is an isomorphism for all p, by induction on B. For $B = 0$, (3.1) gives $H^*_0 = \mathbb{C} = A^0_0$. Let $B \geq 1$ and assume the claim for all smaller B-numbers; equivalently, $\overline{H}^p_b = 0$ for $b < B$. Write $M_p := \ker \psi_{B,p}$.

Degrees $p = 0,1$ and $p = B$. Proposition 8.3 gives $\overline{H}^0_B = \overline{H}^1_B = 0$, i.e. $M_0 = M_1 = 0$, and Lemma 5.2 gives $M_B = 0$.

Separation. Let $B = B_1 + B_2$ with $B_i \geq 1$. By Proposition 6.3 and the induction hypothesis, $K_{B,p} \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2}$ and $A^\vee_{B,p} \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2}$ are isomorphic finitely generated modules over the Noetherian ring $\mathcal{O}^\circ_{B_1,B_2}$. The base change of $\psi_{B,p}$ is a surjection between them; composing it with an abstract isomorphism in the opposite direction gives a surjective endomorphism of a finitely generated module, which is injective [Mat86, Theorem 2.4], [Stacks, Tag 05G8]. Hence the base change of $\psi_{B,p}$ is an isomorphism. By flatness, $M_p \otimes_{\mathcal{O}_B} \mathcal{O}^\circ_{B_1,B_2} = 0$. Now let $\mathfrak{m} \subset \mathcal{O}_B$ be a maximal ideal not containing $\mathfrak{p}_B$, i.e. a monic polynomial of degree B with at least two distinct roots. Factoring it as $f_1 f_2$ with f_1, f_2 coprime of positive degrees gives a point of some U_{B_1,B_2} over $\mathfrak{m}$, at which the local map is flat and local, hence faithfully flat; so $(M_p)_\mathfrak{m} = 0$. Since $\mathcal{O}_B$ is a Jacobson ring, every nonempty locally closed subset of $\operatorname{Spec} \mathcal{O}_B$ contains a maximal ideal; applying this to $\operatorname{Supp} M_p \setminus V(\mathfrak{p}_B)$, we conclude

$$\operatorname{Supp}_{\mathcal{O}_B} M_p \subseteq V(\mathfrak{p}_B) = Z_B.$$

Collision. By Lemma 5.2, $K_{B,p} = H_p(C'_B)$ for $0 \leq p \leq B-1$, where C'_B is a complex of finitely generated free $\mathcal{O}_B$-modules in degrees $0, \ldots, B-1$. Apply Lemma 7.1 with $R = \mathcal{O}_B$, $I = \mathfrak{p}_B$, $c = B-1$, $J_p = A^\vee_{B,p}$ and $C = C'_B$; its depth hypothesis is Lemma 7.2. Hence $M_p = 0$ for $2 \leq p \leq B-1$. (For $B \leq 2$ all degrees were already covered by the previous step.)

Thus every $\psi_{B,p}$ is an isomorphism, which completes the induction. Dualizing each finite-dimensional graded piece shows that φ is bijective. As explained in §2.3, φ is compatible with products, with ∂ and with the λ-brackets, so it is an isomorphism of Poisson vertex algebras. □

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